Project #4

Keytime Animation

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description:

Use keytime animation to make the object move.

The Keytime I declared in the project.

```
// Keytime value
Keytimes Xpos1, Ypos1, Zpos1;
Keytimes Xpos2, Ypos2, Zpos2;
Keytimes Xpos2_L, Ypos2_L, Zpos2_L;
Keytimes Xsun, Ysun, Zsun;
Keytimes ThetaX, ThetaY, ThetaZ;
Keytimes ThetaX2, ThetaY2, ThetaZ2;
Keytimes Red1, Green1, Blue1;
Keytimes Red2, Green2, Blue2;
Keytimes RedL, GreenL, BlueL;
Keytimes RedL2, GreenL2, BlueL2;
Keytimes Xeye1, Yeye1, Zeye1;
Keytimes Xeye2, Yeye2, Zeye2;
// draw the complete scene:
```

4 own-choice animated quantities:

1. Make the airplane's Y axis translates.

```
Ypos1.Init();
Ypos1.AddTimeValue(0.0, 0.000);
Ypos1.AddTimeValue(0.5, 0.318);
Ypos1.AddTimeValue(2.0, 0.800);
Ypos1.AddTimeValue(2.5, 1.318);
Ypos1.AddTimeValue(3.0, 1.518);
Ypos1.AddTimeValue(5.0, 2.200);
Ypos1.AddTimeValue(6.0, 1.710);
Ypos1.AddTimeValue(7.0, 1.310);
Ypos1.AddTimeValue(8.0, 0.610);
Ypos1.AddTimeValue(10.0, 0.000);
```

```
glPushMatrix();
    glTranslatef(Xpos1.GetValue(nowTime), Ypos1.GetValue(nowTime), Zpos1.GetValue(nowTime));
    glRotatef(0., 1., 0., 0.);
    glRotatef(ThetaY.GetValue(nowTime), 0., 1., 0.);
    glRotatef(0., 0., 0., 1.);
//<< draw object #1 >>
    //glTranslatef(0., 0., 0.);
    glRotatef(-180, 0., 1., 0.);
    glRotatef(-90, 1, 0., 0.);
    glScalef(0.0005, 0.0005, 0.0005);

    SetMaterial(0.5 * sin(F_PI * nowTime) +0.5, 0.5 * cos(F_PI* Time) + 0.5, 0.5 * sin(F_PI* Time + 0.25 * F_PI) + 0.5, 100.f);
    glCallList(AirplaneDL);
glPopMatrix();
```

2. Make the airplane's Z axis translates.

```
Zpos1.Init();
Zpos1.AddTimeValue(0.0, 0.000);
Zpos1.AddTimeValue(1.0, 0.718);
Zpos1.AddTimeValue(2.0, 1.500);
Zpos1.AddTimeValue(3.0, 2.018);
Zpos1.AddTimeValue(4.0, 1.318);
Zpos1.AddTimeValue(5.0, 0.000);
Zpos1.AddTimeValue(6.0, -1.000);
Zpos1.AddTimeValue(7.0, -2.000);
Zpos1.AddTimeValue(8.0, -1.500);
Zpos1.AddTimeValue(9.0, -0.500);
Zpos1.AddTimeValue(10.0, 0.000);
```

3. Make the airplane rotate around Y axis.

```
ThetaY.Init();
ThetaY.AddTimeValue(0.0, 0.000);
ThetaY.AddTimeValue(0.5, 0);
ThetaY.AddTimeValue(1.0, 0);
ThetaY.AddTimeValue(2.0, 30);
ThetaY.AddTimeValue(2.5, 70);
ThetaY.AddTimeValue(3.0, 90);
ThetaY.AddTimeValue(3.5, 110);
ThetaY.AddTimeValue(4.0, 130);
ThetaY.AddTimeValue(5.0, 160);
ThetaY.AddTimeValue(6.0, 190);
ThetaY.AddTimeValue(7.0, 230);
ThetaY.AddTimeValue(8.0, 270);
ThetaY.AddTimeValue(9.0, 330);
ThetaY.AddTimeValue(10.0, 360);
```

```
glPushMatrix();
    glTranslatef(Xpos1.GetValue(nowTime), Ypos1.GetValue(nowTime), Zpos1.GetValue(nowTime));
    glRotatef(0., 1., 0., 0.);
    glRotatef(ThetaY.GetValue(nowTime), 0., 1., 0.);
    glRotatef(0., 0., 0., 1.);
//<< draw object #1 >>
    //glTranslatef(0., 0., 0.);
    glRotatef(-180, 0., 1., 0.);
    glRotatef(-90, 1, 0., 0.);
    glScalef(0.0005, 0.0005, 0.0005);

    SetMaterial(0.5 * sin(F_PI * nowTime) +0.5, 0.5 * cos(F_PI* Time) + 0.5, 0.5 * sin(F_PI* Time + 0.25 * F_PI) + 0.5, 100.f);
    glCallList(AirplaneDL);
glPopMatrix();
```

4. Make the color of the light source changes.

```
RedL.Init();
RedL.AddTimeValue(0.0, 0.800);
RedL.AddTimeValue(1.0, 0.600);
RedL.AddTimeValue(2.0, 0.900);
RedL.AddTimeValue(3.0, 0.000);
RedL.AddTimeValue(4.0, 0.800);
RedL.AddTimeValue(5.0, 0.900);
RedL.AddTimeValue(6.0, 1.000);
RedL.AddTimeValue(7.0, 0.700);
RedL.AddTimeValue(8.0, 0.400);
RedL.AddTimeValue(9.0, 0.600);
RedL.AddTimeValue(10.0, 0.900);
```

```
GreenL.Init();
GreenL.AddTimeValue(0.0, 0.000);
GreenL.AddTimeValue(1.0, 1.000);
GreenL.AddTimeValue(2.0, 1.000);
GreenL.AddTimeValue(3.0, 1.000);
GreenL.AddTimeValue(4.0, 0.000);
GreenL.AddTimeValue(5.0, 0.500);
GreenL.AddTimeValue(6.0, 1.000);
GreenL.AddTimeValue(7.0, 0.100);
GreenL.AddTimeValue(8.0, 0.900);
GreenL.AddTimeValue(9.0, 0.000);
GreenL.AddTimeValue(10.0, 1.000);
```

```
    }
    else if (LightColor == WHITE_L && NowLightSource == SPOTLIGHT)
        SetSpotLight(GL_LIGHT0, xlight, ylight, zlight, 0, -0.2, 0, 1., 1., 1.);

    else if (LightColor == CHANGE && NowLightSource == POINTLIGHT)
        SetPointLight(GL_LIGHT0, xlight, ylight, zlight, RedL.GetValue(nowTime), GreenL.GetValue(nowTime), BlueL.GetValue(nowTime));
    else if (LightColor == CHANGE && NowLightSource == SPOTLIGHT)
        SetSpotLight(GL_LIGHT0, xlight, ylight, zlight, 0, -0.2, 0, RedL.GetValue(nowTime), GreenL.GetValue(nowTime), BlueL.GetValue(nowTime));
```

4 required quantities.

1. 1 quantity that has something to do with a light source (position, color, etc.)

   Make the light source moves around.

```
Xsun.Init();
Xsun.AddTimeValue(0.0, 1);
Xsun.AddTimeValue(1.0, 1.3);
Xsun.AddTimeValue(2.0, 1.5);
Xsun.AddTimeValue(3.0, 2.3);
Xsun.AddTimeValue(4.0, 1.4);
Xsun.AddTimeValue(5.0, 0.6);
Xsun.AddTimeValue(6.0, 0.);
Xsun.AddTimeValue(7.0, 0.4);
Xsun.AddTimeValue(8.0, 0.7);
Xsun.AddTimeValue(9.0, 0.9);
Xsun.AddTimeValue(10.0, 1);

Ysun.Init();
Ysun.AddTimeValue(0.0, 2);
Ysun.AddTimeValue(1.0, 1.8);
Ysun.AddTimeValue(2.0, 2.3);
Ysun.AddTimeValue(3.0, 1.7);
Ysun.AddTimeValue(4.0, 2.4);
Ysun.AddTimeValue(5.0, 1.9);
Ysun.AddTimeValue(6.0, 1.8);
Ysun.AddTimeValue(7.0, 1.7);
Ysun.AddTimeValue(8.0, 2.2);
Ysun.AddTimeValue(9.0, 2.6);
Ysun.AddTimeValue(10.0, 2);
```

```
Zsun.Init();
Zsun.AddTimeValue(0.0, 0);
Zsun.AddTimeValue(1.0, 1.3);
Zsun.AddTimeValue(2.0, 1.7);
Zsun.AddTimeValue(3.0, 2.);
Zsun.AddTimeValue(4.0, 1.2);
Zsun.AddTimeValue(5.0, 0.2);
Zsun.AddTimeValue(6.0, -0.9);
Zsun.AddTimeValue(7.0, -1.7);
Zsun.AddTimeValue(8.0, -2.2);
Zsun.AddTimeValue(9.0, -1.3);
Zsun.AddTimeValue(10.0, 0);
```

```
float xlight = Xsun.GetValue(nowTime);
float ylight = Ysun.GetValue(nowTime);
float zlight = Zsun.GetValue(nowTime);

if (LightColor == WHITE_L && NowLightSource == POINTLIGHT)
{
    SetPointLight(GL_LIGHT0, 0., 1.5, zlight, 1., 1., 1.);
}
else if (LightColor == WHITE_L && NowLightSource == SPOTLIGHT)
    SetSpotLight(GL_LIGHT0, xlight, ylight, zlight, 0, -0.2, 0, 1., 1., 1.);
```

2. 1 quantity that has something to do with viewing (eye position, look position, etc.)

   Add a gluLookAt on the airplane, make it looks like a 3rd person mode.

```
// set the eye position, look-at position, and up-vector:
float X_eye1 = Xeye1.GetValue(nowTime);
float Y_eye1 = Yeye1.GetValue(nowTime);
float Z_eye1 = Zeye1.GetValue(nowTime);

float X_eye2 = Xeye2.GetValue(nowTime);
float Y_eye2 = Yeye2.GetValue(nowTime);
float Z_eye2 = Zeye2.GetValue(nowTime);

if( NowLook == OUTSIDE)
    gluLookAt( 4.f, 2.5f, 2.f,     0.f, 0.5f, 0.f,     0.f, 1.f, 0.f );
else if ( NowLook == INSIDE){
    gluLookAt(X_eye1 + 1, Y_eye1 + 1, Z_eye1 + 1, X_eye1, Y_eye1, Z_eye1, 0.f, 1.f, 0.f);
}
else if ( NowLook == CAR)
    gluLookAt(X_eye2 + 1, Y_eye2 + 1, Z_eye2 + 1, X_eye2, Y_eye2, Z_eye2, 0.f, 1.f, 0.f);

// rotate the scene:

glRotatef( (GLfloat)Yrot, 0.f, 1.f, 0.f );
glRotatef( (GLfloat)Xrot, 1.f, 0.f, 0.f );
```

3. 1 quantity that has something to do with positioning, orienting, or scaling the object.

   Make the airplane's X axis translates.

```
Xpos1.Init();
Xpos1.AddTimeValue(0.0, 0.000);
Xpos1.AddTimeValue(0.5, 0.218);
Xpos1.AddTimeValue(1.0, 0.333);
Xpos1.AddTimeValue(2.0, 0.518);
Xpos1.AddTimeValue(3.5, 1.518);
Xpos1.AddTimeValue(4.0, 1.818);
Xpos1.AddTimeValue(5.0, 2.300);
Xpos1.AddTimeValue(7.0, 0.700);
Xpos1.AddTimeValue(9.0, 0.300);
Xpos1.AddTimeValue(10.0, 0.000);
```

4. 1 quantity that has something to do with the color of the object (r, g, or b)
   Change the color of the airplane through time.

```
glPushMatrix();
    glTranslatef(Xpos1.GetValue(nowTime), Ypos1.GetValue(nowTime), Zpos1.GetValue(nowTime));
    glRotatef(0., 1., 0., 0.);
    glRotatef(ThetaY.GetValue(nowTime), 0., 1., 0.);
    glRotatef(0., 0., 0., 1.);
//<< draw object #1 >>
    //glTranslatef(0., 0., 0.);
    glRotatef(-180, 0., 1., 0.);
    glRotatef(-90, 1, 0., 0.);
    glScalef(0.0005, 0.0005, 0.0005);

    SetMaterial( Red1.GetValue(nowTime), Green1.GetValue(nowTime), Blue1.GetValue(nowTime), 100.f);
    glCallList(AirplaneDL);
glPopMatrix();
```

```
Green1.Init();
Green1.AddTimeValue(0.0, 0.000);
Green1.AddTimeValue(1.0, 1.000);
Green1.AddTimeValue(2.0, 0.000);
Green1.AddTimeValue(3.0, 1.000);
Green1.AddTimeValue(4.0, 0.000);
Green1.AddTimeValue(5.0, 1.000);
Green1.AddTimeValue(6.0, 0.000);
Green1.AddTimeValue(7.0, 1.000);
Green1.AddTimeValue(8.0, 0.400);
Green1.AddTimeValue(9.0, 1.000);
Green1.AddTimeValue(10.0, 0.000);
```

```
Blue1.Init();
Blue1.AddTimeValue(0.0, 0.000);
Blue1.AddTimeValue(1.0, 0.300);
Blue1.AddTimeValue(2.0, 0.600);
Blue1.AddTimeValue(3.0, 1.000);
Blue1.AddTimeValue(4.0, 0.700);
Blue1.AddTimeValue(5.0, 0.400);
Blue1.AddTimeValue(6.0, 0.100);
Blue1.AddTimeValue(7.0, 0.300);
Blue1.AddTimeValue(8.0, 0.600);
Blue1.AddTimeValue(9.0, 0.900);
Blue1.AddTimeValue(10.0, 1.000);
```

The plane is flying in circle and can land back on the ground, the color of the plane is changing, the light is moving around and there's a scene that the color of the light is changing.

Extra credit

Description:
Add a bus and make it move.

4 own-choice animated quantities:

1. Make the bus's Y axis translates.

```
Ypos2.Init();
Ypos2.AddTimeValue(0.0, 0.000);
Ypos2.AddTimeValue(1.0, -0.100);
Ypos2.AddTimeValue(2.0, 0.000);
Ypos2.AddTimeValue(3.0, -0.100);
Ypos2.AddTimeValue(4.0, 0.000);
Ypos2.AddTimeValue(5.0, -0.100);
Ypos2.AddTimeValue(6.0, 0.000);
Ypos2.AddTimeValue(7.0, -0.100);
Ypos2.AddTimeValue(8.0, 0.000);
Ypos2.AddTimeValue(9.0, -0.100);
Ypos2.AddTimeValue(10.0, 0.000);
```

2. Make the bus's Z axis translates.

```
Zpos2.Init();
Zpos2.AddTimeValue(0.0, 0.300);
Zpos2.AddTimeValue(0.5, 0.6);
Zpos2.AddTimeValue(1.0, 0.99);
Zpos2.AddTimeValue(2.0, 1.23);
Zpos2.AddTimeValue(3.5, 1.518);
Zpos2.AddTimeValue(4.0, 0.718);
Zpos2.AddTimeValue(4.5, 0.318);
Zpos2.AddTimeValue(5.0, -0.500);
Zpos2.AddTimeValue(5.5, -0.900);
Zpos2.AddTimeValue(6.0, -1.500);
Zpos2.AddTimeValue(7.0, -0.700);
Zpos2.AddTimeValue(9.0, -0.300);
Zpos2.AddTimeValue(10.0, 0.300);
```

3. Make the bus rotate around Y axis.

```
ThetaY2.Init();
ThetaY2.AddTimeValue(0.0, 0);
ThetaY2.AddTimeValue(1.0, 10);
ThetaY2.AddTimeValue(2.0, 60);
ThetaY2.AddTimeValue(3.0, 90);
ThetaY2.AddTimeValue(3.5, 180);
ThetaY2.AddTimeValue(4.0, 200);
ThetaY2.AddTimeValue(4.5, 230);
ThetaY2.AddTimeValue(5.0, 250);
ThetaY2.AddTimeValue(6.0, 270);
ThetaY2.AddTimeValue(7.0, 360);
ThetaY2.AddTimeValue(8.0, 360);
ThetaY2.AddTimeValue(10.0, 360);
```

4. Make the color of the head light of the bus change.

```
BlueL2.Init();
BlueL2.AddTimeValue(0.0, 0.000);
BlueL2.AddTimeValue(1.0, 0.000);
BlueL2.AddTimeValue(2.0, 1.00);
BlueL2.AddTimeValue(3.0, 1.00);
BlueL2.AddTimeValue(4.0, 0.00);
BlueL2.AddTimeValue(5.0, 0.00);
BlueL2.AddTimeValue(6.0, 1.00);
BlueL2.AddTimeValue(7.0, 1.000);
BlueL2.AddTimeValue(8.0, 0.000);
BlueL2.AddTimeValue(9.0, 0.000);
BlueL2.AddTimeValue(10.0, 1.00);
```

```
RedL2.Init();
RedL2.AddTimeValue(0.0, 1.000);
RedL2.AddTimeValue(1.0, 1.000);
RedL2.AddTimeValue(2.0, 0.000);
RedL2.AddTimeValue(3.0, 0.000);
RedL2.AddTimeValue(4.0, 1.000);
RedL2.AddTimeValue(5.0, 1.000);
RedL2.AddTimeValue(6.0, 0.000);
RedL2.AddTimeValue(7.0, 0.000);
RedL2.AddTimeValue(8.0, 1.00);
RedL2.AddTimeValue(9.0, 1.00);
RedL2.AddTimeValue(10.0, 0.000);
```

```
glPushMatrix();
glTranslatef(Xpos2.GetValue(nowTime), Ypos2.GetValue(nowTime), Zpos2.GetValue(nowTime));
glRotatef(ThetaY2.GetValue(nowTime), 0., 1., 0.);
SetSpotLight(GL_LIGHT1,0.3, 0, 0.1, Xpos2_L.GetValue(nowTime), -0.3, abs(Zpos2_L.GetValue(nowTime))
                , RedL2.GetValue(nowTime), 0, BlueL2.GetValue(nowTime));
glPopMatrix();

glShadeModel(GL_SMOOTH);
// uniformly scale the scene:

if( Scale < MINSCALE )
    Scale = MINSCALE;
glScalef( (GLfloat)Scale, (GLfloat)Scale, (GLfloat)Scale );

// set the fog parameters:
```

4 required quantities.

1. 1 quantity that has something to do with a light source (position, color, etc.)

   Make the head light of bus move.

```
glPushMatrix();
glTranslatef(Xpos2.GetValue(nowTime), Ypos2.GetValue(nowTime), Zpos2.GetValue(nowTime));
glRotatef(ThetaY2.GetValue(nowTime), 0., 1., 0.);
SetSpotLight(GL_LIGHT1,0.3, 0, 0.1, Xpos2_L.GetValue(nowTime), -0.3, abs(Zpos2_L.GetValue(nowTime))
                , RedL2.GetValue(nowTime), 0, BlueL2.GetValue(nowTime));
glPopMatrix();

glShadeModel(GL_SMOOTH);
// uniformly scale the scene:

if( Scale < MINSCALE )
    Scale = MINSCALE;
glScalef( (GLfloat)Scale, (GLfloat)Scale, (GLfloat)Scale );

// set the fog parameters:
```

2. 1 quantity that has something to do with viewing (eye position, look position, etc.)

   Add a gluLookAt on the bus, make it looks like a 3rd person mode.

```
// set the eye position, look-at position, and up-vector:
float X_eye1 = Xeye1.GetValue(nowTime);
float Y_eye1 = Yeye1.GetValue(nowTime);
float Z_eye1 = Zeye1.GetValue(nowTime);

float X_eye2 = Xeye2.GetValue(nowTime);
float Y_eye2 = Yeye2.GetValue(nowTime);
float Z_eye2 = Zeye2.GetValue(nowTime);

if( NowLook == OUTSIDE)
    gluLookAt( 4.f, 2.5f, 2.f,     0.f, 0.5f, 0.f,     0.f, 1.f, 0.f );
else if ( NowLook == INSIDE){
    gluLookAt(X_eye1 + 1, Y_eye1 + 1, Z_eye1 + 1, X_eye1, Y_eye1, Z_eye1, 0.f, 1.f, 0.f);
}
else if ( NowLook == CAR)
    gluLookAt(X_eye2 + 1, Y_eye2 + 1, Z_eye2 + 1, X_eye2, Y_eye2, Z_eye2, 0.f, 1.f, 0.f);

// rotate the scene:

glRotatef( (GLfloat)Yrot, 0.f, 1.f, 0.f );
glRotatef( (GLfloat)Xrot, 1.f, 0.f, 0.f );
```

3. 1 quantity that has something to do with positioning, orienting, or scaling the object.

   Make the bus's X axis translates.

```
Xpos2.Init();
Xpos2.AddTimeValue(0.0, 0.300);
Xpos2.AddTimeValue(0.5, 0.3);
Xpos2.AddTimeValue(1.0, 0.3);
Xpos2.AddTimeValue(2.0, 0.518);
Xpos2.AddTimeValue(3.5, 1.418);
Xpos2.AddTimeValue(4.0, 1.618);
Xpos2.AddTimeValue(4.5, 1.718);
Xpos2.AddTimeValue(5.0, 1.500);
Xpos2.AddTimeValue(5.5, 1.200);
Xpos2.AddTimeValue(6.0, 0.600);
Xpos2.AddTimeValue(7.0, 0.300);
Xpos2.AddTimeValue(9.0, 0.300);
Xpos2.AddTimeValue(10.0, 0.300);
```

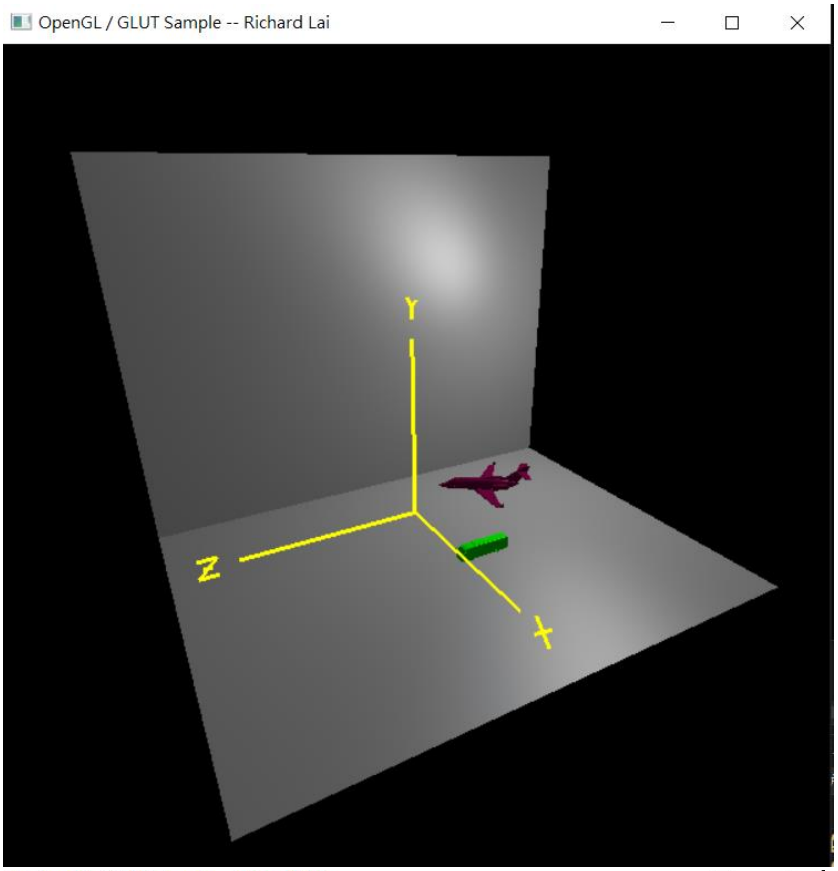
4. 1 quantity that has something to do with the color of the object (r, g, or b)

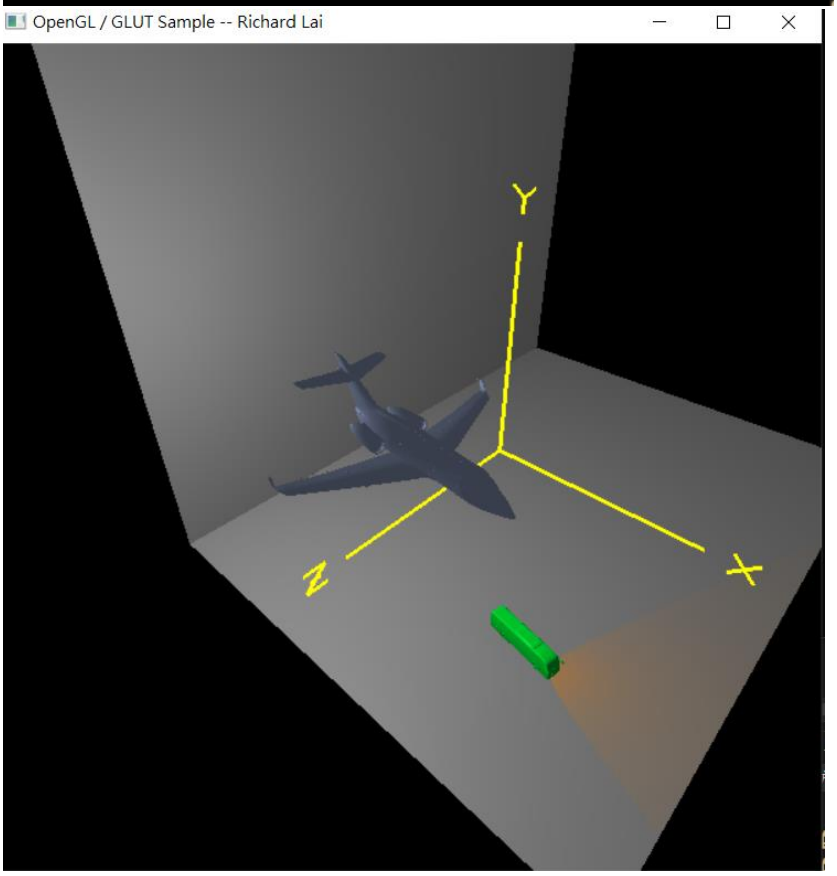
   Change the color of the bus.

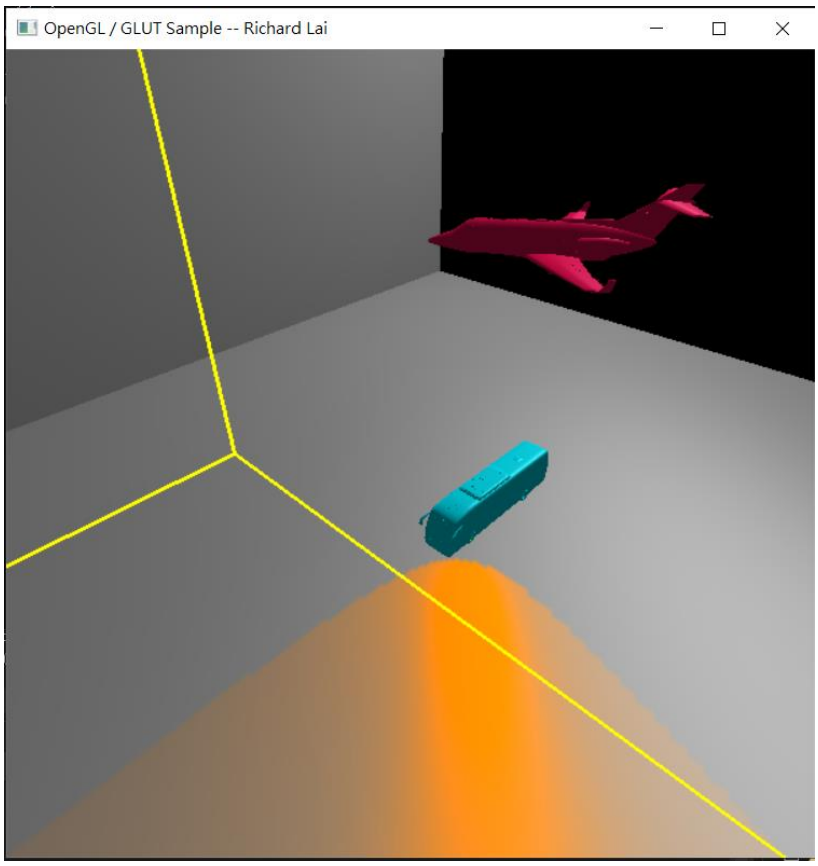
```
glPushMatrix();
glTranslatef(Xpos2.GetValue(nowTime), Ypos2.GetValue(nowTime), Zpos2.GetValue(nowTime));
glRotatef(ThetaY2.GetValue(nowTime), 0., 1., 0.);
glTranslatef(0.3, 0, 0.3);
glRotatef(-180, 0., 1., 0.);
glScalef(0.06, 0.06, 0.06);
SetMaterial(Red2.GetValue(nowTime), Green2.GetValue(nowTime), Blue2.GetValue(nowTime), 100.f);
glCallList(BusDL);
glPopMatrix();
```

The Bus is stopping by the plane when the plane lands on the ground, which seems like it’s loading the passengers that get out of the plane.

Screen shot:


OpenGL / GLUT Sample -- Richard Lai
Y
Z
X


OpenGL / GLUT Sample -- Richard Lai
Y
Z
X



Link : https://media.oregonstate.edu/media/1_2h8t6i8t